

General information

The day-ahead market takes place once per day. Bids are submitted at midday of **day D-1** for the 24 hours of **day D**. Participants must submit their bids separately for each hour.

Each bid comprises:

- The **amount of energy (GWh)** they want to sell or buy
- The **price (€/MWh)** at which they are willing to do so

Market Clearing Process

For each hour, generation bids are ordered from the lowest to the highest price, forming the **supply curve**, while demand bids are ordered from the highest to the lowest price, forming the **demand curve**. The intersection of these two curves determines the **market clearing price**, the amount of **energy traded**, and the **generation schedule** for that hour.

Uniform-Price Auction

The day-ahead market is a **uniform-price** auction, which means that all accepted generators are paid the same price and all buyers pay the same price. This price corresponds to the market clearing price. In theory, generators maximise their profits by bidding their opportunity cost (that is, their avoidable production cost), since this increases the probability of being dispatched while ensuring that their costs are covered.

Imbalances

The day-ahead market fixes revenues and payments **one day in advance (D-1)**. However, real production and consumption during **day D** may differ from scheduled values for several reasons. These deviations are called **imbalances**, and they lead to **penalties**.

Game Description

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Public information

There are five generators, each representing one technology:

- Nuclear: 8 GW
- Gas: 20 GW
- Coal: 10 GW
- Hydro: 15 GW
- Solar-Wind: 20 GW

At the beginning of each round, the Market Operator will provide an **estimate of total demand** for that hour. In addition, relevant market information will be published, including the **hydro reserves**, the **CO₂ price**, and any other data necessary to support your bidding decisions.

Participant: COAL

You are responsible for the **COAL** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

- The **price** per megawatt-hour (€/MWh) at which you are willing to sell your energy. The minimum value (floor) is 0 €/MWh. The maximum value (cap) is 180 €/MWh.
- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 10 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **30 €/MWh**.
2. Your CO₂ emissions are **0.8 tCO₂/MWh** (only if the ETS market is active).
3. Since the power output of coal plants can be adjusted, you won't incur penalties. If you sell a certain amount of energy in the market, you will deliver exactly the same amount of energy at the corresponding hour.

For example, if you sell 8 GWh in the market, and the market clearing price is 40 €/MWh:

- You earn $8 \times 40 = 320$ thousand € (market incomes).
- You pay $8 \times 30 = 240$ thousand € for the fuel consumption (variable cost).
- Thus, overall, you earn $320 - 240 = 80$ thousand € (total incomes).

Additionally, if the ETS market were active and the CO₂ price were 10 €/tCO₂, your variable costs would increase by $10 \times 0.8 \times 8 = 64$ thousand €.

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Participant: GAS

You are responsible for the **GAS** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

- The **price** per megawatt-hour (€/MWh) at which you are willing to sell your energy. The minimum value (floor) is 0 €/MWh. The maximum value (cap) is 180 €/MWh.
- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 20 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **50 €/MWh**.
2. Your CO₂ emissions are **0.4 tCO₂/MWh** (only if the ETS market is active).
3. Since the power output of gas plants can be adjusted, you won't incur penalties. If you sell a certain amount of energy in the market, you will deliver exactly the same amount of energy at the corresponding hour.

For example, if you sell 8 GWh in the market, and the market clearing price is 60 €/MWh:

- You earn $8 \times 60 = 480$ thousand € (market incomes).
- You pay $8 \times 50 = 400$ thousand € for the fuel consumption (variable cost).
- Thus, overall, you earn $480 - 400 = 80$ thousand € (total incomes).

Additionally, if the ETS market were active and the CO₂ price were 10 €/tCO₂, your variable costs would increase by $10 \times 0.4 \times 8 = 32$ thousand €.

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Participant: HYDRO

You are responsible for the **HYDRO** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

- The **price** per megawatt-hour (€/MWh) at which you are willing to sell your energy. The minimum value (floor) is 0 €/MWh. The maximum value (cap) is 180 €/MWh.
- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 15 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **0 €/MWh**.
2. You don't have CO₂ emissions associated with generation.
3. Since the power output of hydro plants can be adjusted, in general you won't incur penalties. If you sell a certain amount of energy in the market, you will deliver exactly the same amount of energy at the corresponding hour.
4. However, the hydro reserves have a limited amount of water. You can only generate a total of **30 GWh** in four rounds. If you exceed this limit during the game by selling more energy than you can generate, you will incur a penalty according to the energy imbalance. The penalty price is different for each round, and depends on the overall imbalance, considering deviations from all participants.

For example, imagine that this is Round 4 and that you have already generated 23 GWh in previous rounds. Therefore, you only have enough water to generate 7 GWh. If you sell 10 GWh in the market, you create an imbalance of 3 GWh. If the market clearing price is 50 €/MWh:

- You earn $10 \times 50 = 500$ thousand € (**market incomes**).
- If the penalty price were 70 €/MWh, you would pay a **penalty** of $3 \times 70 = 210$ thousand €.
- Thus, overall, you earn $500 - 210 = 290$ thousand € (**total incomes**).

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Participant: NUCLEAR

You are responsible for the **NUCLEAR** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

- The **price** per megawatt-hour (€/MWh) at which you are willing to sell your energy. The minimum value (floor) is 0 €/MWh. The maximum value (cap) is 180 €/MWh.
- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 8 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **5 €/MWh**.
2. You don't have CO₂ emissions associated with generation.
3. Nuclear plants must work at full capacity. This means that, regardless the energy you sell in each round, your plant will deliver 10 GWh. If you sell less than that, you will pay a penalty (because you will inject more energy than that compromised in the market, thus you create an imbalance). The penalty price is different for each round, and depends on the overall imbalance, considering deviations from all participants.

For example, if you sell 6 GWh in the market, since you actually always generate 8 GWh, you create an imbalance of 2 GWh. If the market clearing price is 20 €/MWh:

- You earn $6 \times 20 = 120$ thousand € (market incomes).
- You pay $8 \times 5 = 40$ thousand € for the fuel consumption (variable cost).
- If the penalty price were 30 €/MWh, you would pay a penalty of $2 \times 30 = 60$ thousand €.
- Thus, overall, you earn $120 - 40 - 60 = 20$ thousand € (total incomes).

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Participant: SOLAR-WIND

You are responsible for the **SOLAR-WIND** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

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- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 20 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **0 €/MWh**.
2. You don't have CO₂ emissions associated with generation.
3. Solar-Wind generation can be adjusted downward (i.e. you can curtail the power output), but not upward (your generation is limited by the available resource). Therefore, accurate forecasts are important for setting your bids and reducing imbalances. If you sell more energy in the market (at day *D-1*) than you actually generate at day *D*, you will incur a penalty according to the energy imbalance. The penalty price is different for each round, and depends on the overall imbalance, considering deviations from all participants.

For example, if you sell 12 GWh in the market but you generate only 10 GWh, you create an imbalance of 2 GWh. If the market clearing price is 20 €/MWh:

- You earn $12 \times 20 = 240$ thousand € (market incomes).
- If the penalty price were 30 €/MWh, you would pay a penalty of $2 \times 30 = 60$ thousand €.
- Thus, overall, you earn $240 - 60 = 180$ thousand € (total incomes).

Day 1

You hired a new forecasting company with little experience in the field, but cheap prices.

They work with point-forecasting models

They provided the following power forecast for your solar farm portfolio:

11 GWh

Day 2

To improve your bids, you hired four forecasting companies.

All four generate point-forecasts.

They provided the following power forecasts for your solar farm portfolio:

15 GWh

17 GWh

18 GWh

18 GWh

Day 3

You are still working with these four companies.

They provided the following power forecasts for your solar farm portfolio:

6 GWh

7 GWh

18 GWh

20 GWh

When asked about these discrepancies, they inform you that a weather front is approaching the Iberian peninsula, and is expected to arrive by 2 p.m. tomorrow. Two companies predict it will arrive slightly earlier, resulting in a low solar PV generation due to clouds. The other two companies predict a delay in the arrival of the weather front, which would result in high solar generation.

Day 4

You hired a top-tier company that works with probabilistic forecasting models.

The situation is similar to that of the previous round: A weather front is approaching the Iberian Peninsula, so solar power generation is highly volatile.

The following forecast of the solar power probability density function has been reported for your solar farm portfolio:

